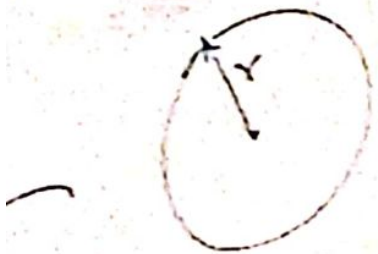
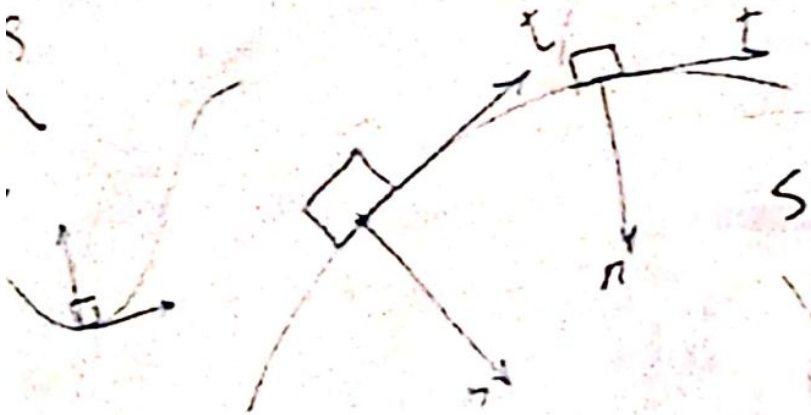
$\rho \rightarrow$ radius of curvature





$S = 10t^2 + 2t + 5$

$V = \frac{ds}{dt}$

x	y
V_x	V_y
a_x	a_y

$V = \sqrt{V_x^2 + \cdots}$

$a = \sqrt{a_x^2 + \cdots}$

t	n
$V_t = \frac{ds}{dt}$	$V_n = 0$
$a_t = \frac{dv}{dt}$	$a_n = V^2/\rho$

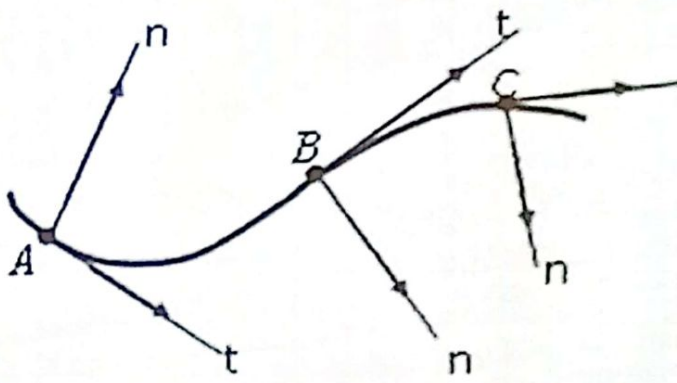
$V = \sqrt{V_t^2 + V_n^2}$

$a = \sqrt{a_t^2 + \cdots}$

r	θ
V_r	V_θ
a_r	a_θ

Curvilinear Motion : Normal And Tangential Coordinates

One of the common descriptions of curvilinear motion involves the use of path variables, which are measurements made along the tangent t and normal n to the path of the particle. These coordinates provide a natural description for curvilinear motion and are direct and convenient coordinates to use. The n- and t-coordinates are considered to move along the path with the particle as seen where the particle advances from A to B to C.



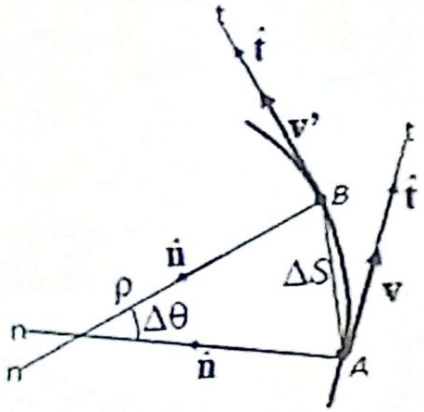
Curvilinear Motion : Normal And Tangential Coordinates

The magnitude of the *average velocity* may be written as

$$\Delta v = \frac{\Delta s}{\Delta t} = \rho \frac{\Delta \theta}{\Delta t}$$

As $\Delta t \to 0$ the magnitude of the instantaneous velocity becomes

$$v = \frac{dS}{dt} = \dot{S}$$

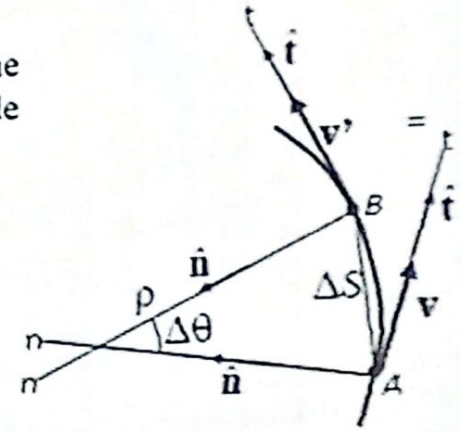


Curvilinear Motion : Normal And Tangential Coordinates

the magnitude of the velocity is equal to the derivative of the path function, $s(t)$. An equivalent expression for the magnitude of the velocity is given by

$$v = \rho\frac{d\theta}{dt} = \rho\dot{\theta}$$

$\dot{\theta}$ is known as the angular velocity and equals the rate of change of the angle θ with respect to the time t.



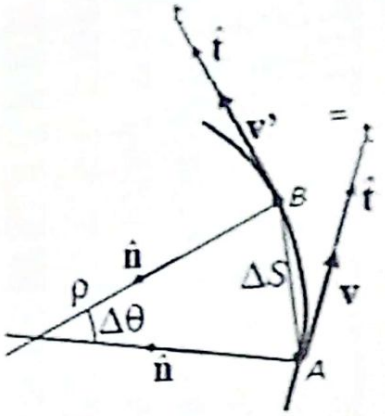
Curvilinear Motion : Normal And Tangential Coordinates

The velocity is always tangent to the curve. Thus we may write the velocity vector as

$$v = \dot{S}\,\hat{t}$$

or

$$v = \rho\dot{\theta}\,\hat{t}$$

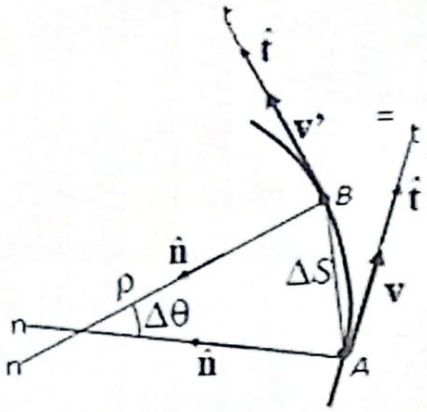


The acceleration a of the particle is defined by

$$\mathbf{a} = \frac{dv}{dt} = \frac{d}{dt}(v\,\hat{\mathbf{t}}) = \frac{dv}{dt}\hat{\mathbf{t}} + v\frac{d\hat{\mathbf{t}}}{dt}$$

Curvilinear Motion : Normal And Tangential Coordinates

$$a = a_n \hat{\mathbf{n}} + a_t \hat{\mathbf{t}}$$



The *normal component of acceleration* is

$$a_n = \frac{v^2}{\rho}.$$

The *tangential component of acceleration* is

$$a_t = \frac{dv}{dt} = \dot{v}.$$

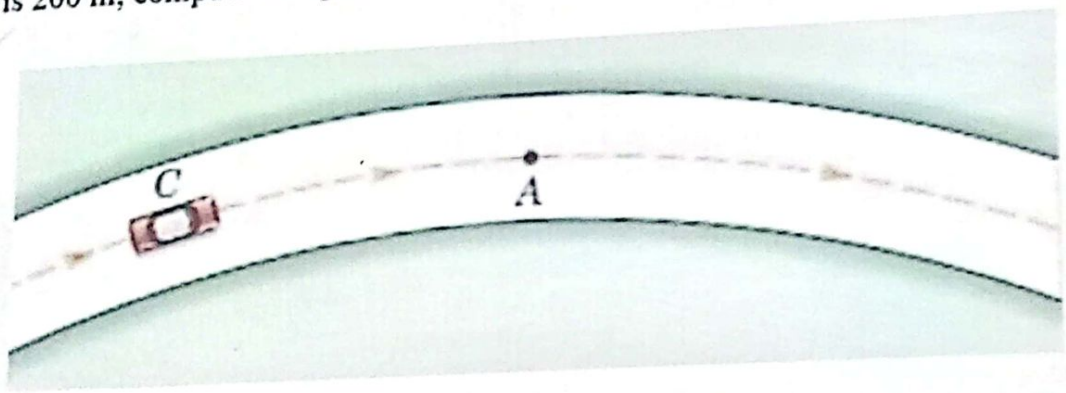
The magnitude of the acceleration is given by

$$a = \sqrt{a_n^2 + a_t^2}.$$

CamScanner

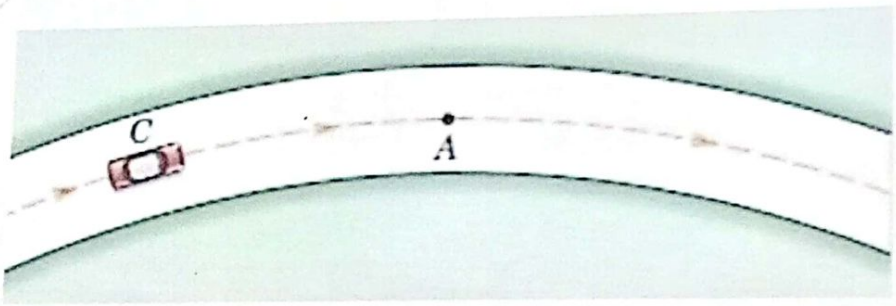
Curvilinear Motion : Normal And Tangential Coordinates

Ex:- The car C increases its speed at the constant rate of $1.6\ m/s^2$ as it rounds the curve shown. If the magnitude of the total acceleration of the car is $2.8\ m/s^2$ at point A where the radius of curvature is 200 m, compute the speed v of the car at this point.



Curvilinear Motion :
Normal And Tangential Coordinates

Ex:- The car C increases its speed at the **constant rate of 1.6 m/s²** as it rounds the curve shown. If the magnitude of **the total acceleration of the car is 2.8 m/s²** at point A where **the radius of curvature is 200 m,** compute the speed v of the car at this point.



Curvilinear Motion : Normal And Tangential Coordinates

Ex:- The car C increases its speed at the **constant rate of 1.6 m/s²** as it rounds the curve shown. If the magnitude of **the total acceleration of the car is 2.8 m/s²** at point A where **the radius of curvature is 200 m,** compute the speed v of the car at this point.

$$a_t = 1.6 \text{ m/s}^2 .$$

$$a = 2.8 \text{ m/s}^2$$

$$\rho = 200 \text{ m.}$$

CamScanner

Curvilinear Motion : Normal And Tangential Coordinates

$a_t = 1.6$ m/s^2 .

$a = 2.8$ m/s^2

$\rho = 200$ m.

the normal component of acceleration is given by

$$a_n = \sqrt{a^2 - a_t^2}$$

$$= \sqrt{(2.8)^2 - (1.6)^2} = 2.298 \text{ m/s}^2$$

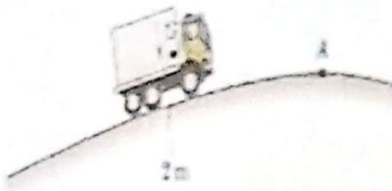
e speed is obtained as follows:

$$v = \sqrt{a_n \rho}$$

$$= \sqrt{(2.298)(200)} = 21.44 \text{ m/s.}$$

Curvilinear Motion : Normal And Tangential Coordinates

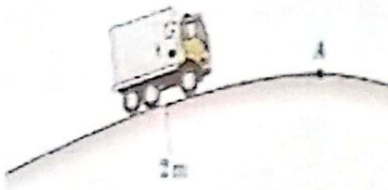
Ex:- The driver of the truck has an acceleration of 0.4g as the truck passes over the top A of the hump in the road at constant speed. The radius of curvature of the road at the top of the hump is 98 m, and the center of mass G of the driver (considered as a particle) is 2 m above the road. Calculate the speed v of the truck.



CamScanner

Curvilinear Motion :
Normal And Tangential Coordinates

Ex:- The driver of the truck has an **acceleration of 0.4g** as the truck passes over the top A of the hump in the road at **constant speed.** The radius of curvature of the road at the top of the hump is 98 m, and the center of mass G of the driver (considered as a particle) is 2 m above the road. Calculate the speed v of the truck.

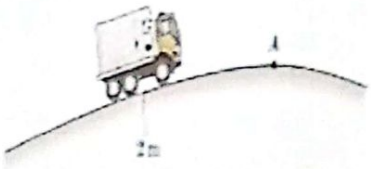


CamScanner

Curvilinear Motion : Normal And Tangential Coordinates

The radius of curvature to the center of mas:

$$\rho = 98 + 2 = 100 \text{ m}$$



Since the truck speed is constant, then

$$a_t = \frac{dv}{dt} = 0$$

The acceleration vector is then,

$$\boldsymbol{a} = a_n \hat{\mathbf{n}} + a_t \hat{\mathbf{t}} = a_n \hat{\mathbf{n}}$$

i.e., the magnitude of the acceleration is $a = |\boldsymbol{a}| = a_n = 0.4$ g.

Thus,

$$\frac{v^2}{\rho} = 0.4\ g.$$

yielding the truck speed as

$$v = \sqrt{40g} = 19.8 \text{ m/s}.$$

Curvilinear Motion : Normal And Tangential Coordinc



Ex:- The muzzle velocity for a certain rifle is 600 m/s. If the rifle is pointed vertically upward and fired from an automobile moving horizontally at a speed of 72 km/h, determine the radius of curvature of the path of the bullet at its maximum altitude. Neglect air resistance.

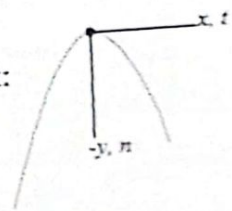
CamScanner

Curvilinear Motion :
Normal And Tangential Coordinates

Ex:- The muzzle velocity for a certain rifle is 600 m/s. If the rifle is pointed vertically upward and fired from an automobile moving horizontally at a speed of 72 km/h, determine the radius of curvature of the path of the bullet at its maximum altitude. Neglect air resistance.

We note the following:

1. At maximum altitude, the vertical component of the bullet velocity is zero.
2. Once in the air, the bullet is a projectile in free motion. (air resistance is neglected)



From the motion of projectile, we know that:

$a_x = 0$, and
$a_y = -9.8$ m/s^2 (downwards)

CamScanner

Curvilinear Motion : Normal And Tangential Coordinates

Ex:- The muzzle velocity for a certain rifle is 600 m/s. If the rifle is pointed vertically upward and fired from an automobile moving horizontally at a speed of 72 km/h, determine the radius of curvature of the path of the bullet at its maximum altitude. Neglect air resistance.

$a_t = a_x = 0$ and
$a_n = -a_y = 9.8\ m/s^2$

$v_y = 0$ $\quad v_x = 72\ km/h = (72/3.6)\ m/s = 20\ m/s$

$v = \sqrt{v_x^2 + v_y^2} = v_x = 20\ m/s$

$$a_n = \frac{v^2}{\rho} \qquad 9.8 = \frac{(20)^2}{\rho}$$

$$\rho = 40.8\ m$$